

This page covers the shift from a naive Poisson distribution to a **Poisson regression model (Generalized Linear Model)**.

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1. The Core Problem It Solves

A pure Poisson model assumes a single, static baseline rate λ across all conditions (e.g., $\lambda=5$).

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In the physical world, event rates are never constant:

- Bike share usage spikes during rush hours and warm weather.
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- In enterprise security, baseline motion/crossing events vary drastically by hour of the day, day of the week, shift changes, weather (wind/rain blowing foliage), and operational sector status.
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Rather than treating the mean $\lambda=E(Y)$ as a constant scalar, **Poisson regression models λ as a function of environmental covariates $X_1, \dots, X_p$:**

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$$\lambda(X)=E(Y|X)$$

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2. The Log Link Function (Equations 4.36 & 4.37)

Linear regression assumes $E(Y|X)=\beta_0+\beta_1X_1+\dots+\beta_pX_p$. That fails for event rates because a linear combination can easily evaluate to a negative number ($\lambda<0$), which is mathematically impossible for an event rate or count.

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To enforce $\lambda(X)>0$ under all conditions, the model uses the **log link function**:

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$$\log(\lambda(X_1, \dots, X_p))=\beta_0+\beta_1X_1+\dots+\beta_pX_p$$

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Exponentiating both sides yields:

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$$\lambda(X_1, \dots, X_p)=e^{\beta_0+\beta_1X_1+\dots+\beta_pX_p}=\exp(\beta^T X)$$

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Because the natural exponential function $e^z > 0$ for all real z , **the expected event rate $\lambda(\mathbf{X})$ is guaranteed to stay strictly positive** regardless of the feature values or estimated weights.

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3. Estimating Parameters via Maximum Likelihood (Equation 4.38)

Unlike standard linear regression (which minimizes squared residuals using Ordinary Least Squares), Poisson regression parameters $\beta_0, \beta_1, \dots, \beta_p$ are estimated using **Maximum Likelihood Estimation (MLE)**.

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For n independent observations $(x_1, y_1), \dots, (x_n, y_n)$, the likelihood function is the product of the individual Poisson probabilities:

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$$\ell(\beta_0, \beta_1, \dots, \beta_p) = \prod_{i=1}^n \frac{e^{-\lambda(x_i)} [\lambda(x_i)]^{y_i}}{y_i!}$$

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Taking the log gives the **log-likelihood** $L(\beta)$, which turns the product into a sum:

$$L(\beta) = \sum_{i=1}^n (y_i \log(\lambda(x_i)) - \lambda(x_i) - \log(y_i!))$$

Substituting $\log(\lambda(x_i)) = \beta^T x_i$:

$$L(\beta) = \sum_{i=1}^n (y_i (\beta^T x_i) - e^{\beta^T x_i} - \log(y_i!))$$

Because $L(\beta)$ is globally concave, it converges rapidly and deterministically using standard Newton-Raphson or Iteratively Reweighted Least Squares (IRLS).

4. Direct Connection to VectorEdge

This exact formula explains why dynamic baseline modeling can execute on a low-cost, low-power edge processor without running heavy neural networks:

1. **Offline/Cloud Calibration or Nightly Recalibration:** The coefficients $\beta = [\beta_0, \beta_1, \dots, \beta_p]$ are fit using historical telemetry logs. The vector β is just a handful of floating-point scalars (e.g., weights for `hour_sin`, `hour_cos`, `ambient_lux`, `wind_speed`, `sector_id`).
2. **Microsecond Runtime Inference on \$15 Edge Silicon:** At runtime, the edge daemon does not run a matrix-heavy deep net. To compute the expected baseline rate λ_{nominal} for the current 5-minute window, it computes a simple dot product and one scalar exponentiation:

$$\lambda_{\text{current}} = \exp(\beta_0 + \sum_{j=1}^p \beta_j X_j)$$

- This executes in single-digit microseconds on an ARM Cortex core.
- 3. **Autonomous Statistical Thresholding:** Once λ_{current} is known, an observed burst of k crossings (whether hamsters, vehicles, or intruders) is scored against the exact probability mass function:

$$\Pr(Y \geq k \mid \lambda_{\text{current}})$$

If $p < \alpha_{\text{threshold}}$, the event is statistically certified as an anomaly and triggers a 64-byte alert payload; if $p \geq \alpha_{\text{threshold}}$, it is treated as nominal environmental drift and suppressed entirely.